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Integral Curves and Flows

1.1 Integral Curves

Definition 1.1.1 ► Integral Curve

If V is a vector field on M , an **integral curve of V** is a differentiable curve $\gamma : J \rightarrow M$ whose velocity at each point is equal to the value of V at that point:

$$\gamma'(t) = V_{\gamma(t)} \quad \text{for all } t \in J.$$

If $0 \in J$, the point $\gamma(0)$ is called the **starting point of γ** .

Finding integral curves boils down to solving a system of ordinary differential equations in a smooth chart. Suppose V is a smooth vector field on M and $\gamma : J \rightarrow M$ is a smooth curve. On a smooth coordinate domain $U \subseteq M$, we can write γ in local coordinates as

$$\gamma(t) = (\gamma^1(t), \dots, \gamma^n(t))$$

Then the condition $\gamma'(t) = V_{\gamma(t)}$ for γ to be an integral curve of V can be written

$$\dot{\gamma}^i(t) \frac{\partial}{\partial x^i} \Big|_{\gamma(t)} = V^i(\gamma(t)) \frac{\partial}{\partial x^i} \Big|_{\gamma(t)},$$

which reduces to the following autonomous system of ordinary differential equations:

$$\begin{aligned} \dot{\gamma}^1(t) &= V^1(\gamma^1(t), \dots, \gamma^n(t)), \\ &\vdots \\ \dot{\gamma}^n(t) &= V^n(\gamma^1(t), \dots, \gamma^n(t)). \end{aligned}$$

Proposition 1.1.2

Let V be a smooth vector field on a smooth manifold M . For each point $p \in M$, there exists $\varepsilon > 0$ and a smooth curve $\gamma : (-\varepsilon, \varepsilon) \rightarrow M$ that is an integral curve of V starting at p .

Proof. The ODE system corresponding to the condition

$$\gamma'(t) = V_{\gamma(t)}$$

is smooth. Thus it satisfies the local Lipschitz condition. Then the proposition follows from the Picard-Lindelöf Theorem. $\square$

Proposition 1.1.3 ► Naturality of Integral Curves

Suppose M and N are smooth manifolds and $F : M \rightarrow N$ is a smooth map. Then $X \in \mathfrak{X}(M)$ and $Y \in \mathfrak{X}(N)$ are F -related if and only if F takes integral curves of X to integral curves of Y , meaning that for each integral curve γ of X , $F \circ \gamma$ is an integral curve of Y .

Proof. Suppose that X and Y are F -related, and $\gamma : J \rightarrow M$ is an integral curve of X . If we define $\sigma : J \rightarrow N$ by $\sigma = F \circ \gamma$, then

$$\sigma'(t) = dF_{\gamma(t)}\gamma'(t) = dF_{\gamma(t)}X_{\gamma(t)} = Y_{F(\gamma(t))} = Y_{\sigma(t)}$$

so σ is an integral curve of Y .

Conversely, suppose F take integral curves of X to integral curves of Y . Given $p \in M$, let $\gamma : (-\varepsilon, \varepsilon) \rightarrow M$ be an integral curve of X starting at p . Since $F \circ \gamma$ is an integral curve of Y starting at $F(p)$, we have

$$Y_{F(p)} = (F \circ \gamma)'(0) = dF_p(\gamma'(0)) = dF_p(X_p)$$

which shows that X and Y are F -related. $\square$

1.2 Flows

Definition 1.2.1

We define a **global flow** on M (also called a **one-parameter group action**) to be a continuous left $\mathbb{R}$ -action on M ; that is, a continuous map $\theta : \mathbb{R} \times M \rightarrow M$ satisfying the following properties for all $s, t \in \mathbb{R}$ and $p \in M$:

$$\theta(t, \theta(s, p)) = \theta(t + s, p), \quad \theta(0, p) = p. \quad (1.1)$$

Given a global flow θ on M , we define two collections of maps as follows:

- For each $t \in \mathbb{R}$, define a continuous map $\theta_t : M \rightarrow M$ by

$$\theta_t(p) = \theta(t, p).$$

The defining properties (1.1) are equivalent to the **group laws**

$$\theta_t \circ \theta_s = \theta_{t+s}, \quad \theta_0 = \text{Id}_M. \quad (1.2)$$

As is the case for any continuous group action, each map $\theta_t : M \rightarrow M$ is a homeomorphism, and if the flow is smooth, θ_t is a diffeomorphism.

- For each $p \in M$, define a curve $\theta^{(p)} : \mathbb{R} \rightarrow M$ by

$$\theta^{(p)}(t) = \theta(t, p).$$

The image of this curve is the orbit of p under the group action.

Proposition 1.2.2

If $\theta : \mathbb{R} \times M \rightarrow M$ is a smooth global flow, for each $p \in M$ we define a tangent vector $V_p \in T_p M$ by

$$V_p = \theta^{(p)'}(0).$$

The assignment $p \mapsto V_p$ is a (rough) vector field on M , which is called the **infinitesimal generator of θ** .

The infinitesimal generator V of θ is a smooth vector field on M , and each curve $\theta^{(p)}$ is an integral curve of V .

1.2.1 Fundamental Theorem on Flows

Definition 1.2.3

- If M is a manifold, a **flow domain** for M is an open subset $\mathcal{D} \subseteq \mathbb{R} \times M$ with the property that for each $p \in M$, the set $\mathcal{D}^{(p)} = \{t \in \mathbb{R} : (t, p) \in \mathcal{D}\}$ is an open interval containing 0.
- A **flow** on M is a continuous map $\theta : \mathcal{D} \rightarrow M$, where $\mathcal{D} \subseteq \mathbb{R} \times M$ is a flow domain, that satisfies the following group laws :
 - For all $p \in M$,

$$\theta(0, p) = p, \quad (1.3)$$

– and for all $s \in \mathcal{D}^{(p)}$ and $t \in \mathcal{D}^{(\theta(s,p))}$ such that $s + t \in \mathcal{D}^{(p)}$,

$$\theta(t, \theta(s, p)) = \theta(t + s, p). \quad (1.4)$$

We sometimes call θ a **local flow** to distinguish it from a global flow as defined earlier. The unwieldy term **local one-parameter group action** is also used.

Note.

- **flows domain** 要求每一个空间节点上, 时间的变化范围都是包含 0 的开区间
- 一个 **flow** 就是 M 上活动在 $\mathcal{D}$ 内关于时间变化的一个局部左 $\mathbb{R}$ -作用.

Definition 1.2.4

If θ is a flow, we define $\theta_t(p) = \theta^{(p)}(t) = \theta(t, p)$ whenever $(t, p) \in \mathcal{D}$, just as for a global flow. For each $t \in \mathbb{R}$, we also define

$$M_t = \{p \in M : (t, p) \in \mathcal{D}\}, \quad (1.5)$$

so that

$$p \in M_t \Leftrightarrow t \in \mathcal{D}^{(p)} \Leftrightarrow (t, p) \in \mathcal{D}.$$

If θ is smooth, the **infinitesimal generator of θ** is defined by $V_p = \theta^{(p)'}(0)$.

Note.

- 当我们谈论点在 t 时间后的行为时, 只能对 M_t 上的点讨论.
- 无穷小生成元就是每一点处运动曲线的起始切向量.

Theorem 1.2.5 ► Fundamental Theorem on Flows

Let V be a smooth vector field on a smooth manifold M . There is a unique smooth maximal flow $\theta : \mathcal{D} \rightarrow M$ whose infinitesimal generator is V . This flow has the following properties:

- For each $p \in M$, the curve $\theta^{(p)} : \mathcal{D}^{(p)} \rightarrow M$ is the unique maximal integral curve of V starting at p .
- If $s \in \mathcal{D}^{(p)}$, then $\mathcal{D}^{(\theta(s,p))}$ is the interval $\mathcal{D}^{(p)} - s = \{t - s : t \in \mathcal{D}^{(p)}\}$.
- For each $t \in \mathbb{R}$, the set M_t is open in M , and $\theta_t : M_t \rightarrow M_{-t}$ is a diffeomorphism with inverse θ_{-t} .

Note.

- 向量场 V 的最大流 $\theta : \mathcal{D} \subset \mathbb{R} \times M \rightarrow M$ 将从 M 中各点出发的极大积分曲线统一表示为 $\gamma_p(t) = \theta(t, p)$. 相对于单纯的一族积分曲线, 最大流 θ 还同时编码了解对初值 p 的光滑依赖, 即 θ 作为 (t, p) 的函数是光滑的
- 直观上, [(c)] 用平移来看. 划定一个区域, 将这些区域整体平移, 删去那些跑到区域外的点, 删去的点正好能补齐因为平移而产生的空缺, 这体现了平移关于时间的可逆性, 平移就是一种特殊的流.

1.2.2 Complete Vector Fields

Definition 1.2.6

We say that a smooth vector field is **complete** if it generates a global flow, or equivalently if each of its maximal integral curves is defined for all $t \in \mathbb{R}$.

Lemma 1.2.7 ▶ Uniform Time Lemma

Let V be a smooth vector field on a smooth manifold M , and let θ be its flow. Suppose there is a positive number ε such that for every $p \in M$, the domain of $\theta^{(p)}$ contains $(-\varepsilon, \varepsilon)$. Then V is complete.

Proof. Suppose for the sake of contradiction for some $p \in M$, the domain $\mathcal{D}^{(p)}$ of $\theta^{(p)}$ is bounded above. Let $b = \sup \mathcal{D}^{(p)}$, let t_0 be a positive number such that $b - \varepsilon < t_0 < b$, and let $q = \theta^{(p)}(t_0)$. The hypothesis implies that $\theta^{(q)}(t)$ is defined at least for $t \in (-\varepsilon, \varepsilon)$. Define curve $\gamma : (-\varepsilon, t_0 + \varepsilon) \rightarrow M$ by

$$\gamma(t) = \begin{cases} \theta^{(p)}(t), & -\varepsilon < t < b \\ \theta^{(q)}(t - t_0), & t_0 - \varepsilon < t < t_0 + \varepsilon \end{cases}$$

These two definitions agree where they overlap, because $\theta^{(q)}(t - t_0) = \theta_{t-t_0}(q) = \theta_{t-t_0}\theta_{t_0}(p) = \theta_t(p) = \theta^{(p)}(t)$ by the group law for θ . γ is an integral curve starting at p . Since $t_0 + \varepsilon > b$, this is a contradiction. $\square$

1.3 Lie Derivatives

Definition 1.3.1

Suppose M is a smooth manifold, V is a smooth vector field on M , and θ is the flow of V . For any smooth vector field W on M , define a rough vector field on M , denoted by $\mathcal{L}_V W$ and called the **Lie derivative of W with respect to V** , by

$$\begin{aligned} (\mathcal{L}_V W)_p &= \left. \frac{d}{dt} \right|_{t=0} d(\theta_{-t})_{\theta_t(p)}(W_{\theta_t(p)}) \\ &= \lim_{t \rightarrow 0} \frac{d(\theta_{-t})_{\theta_t(p)}(W_{\theta_t(p)}) - W_p}{t}, \end{aligned} \quad (1.6)$$

provided the derivative exists. For small $t \neq 0$, at least the difference quotient makes sense: θ_t is defined in a neighborhood of p , and θ_{-t} is the inverse of θ_t , so both $d(\theta_{-t})_{\theta_t(p)}(W_{\theta_t(p)})$ and W_p are elements of $T_p M$.

Theorem 1.3.2

If M is a smooth manifold and $V, W \in \mathfrak{X}(M)$, then $\mathcal{L}_V W = [V, W]$.

1.4 Commuting Vector Fields

Definition 1.4.1

If θ is a smooth flow, a vector field W is said to be **invariant under θ** if W is θ_t -related to itself for each t ; more precisely, this means that $W|_{M_t}$ is θ_t -related to $W|_{M_{-t}}$ for each t , or equivalently that $d(\theta_t)_p(W_p) = W_{\theta_t(p)}$ for all (t, p) in the domain of θ .

Theorem 1.4.2

For smooth vector fields V and W on a smooth manifold M , the following are equivalent:

- (a) V and W commute.
- (b) W is **invariant under the flow** of V .
- (c) V is **invariant under the flow** of W .

Theorem 1.4.3 ► Canonical Form for Commuting Vector Fields

Let M be a smooth n -manifold, and let $(V_1, \dots, V_k)$ be a linearly independent k -tuple of smooth commuting vector fields on an open subset $W \subseteq M$.

- For each $p \in W$, there exists a smooth coordinate chart $(U, (s^i))$ centered at p such that $V_i = \partial/\partial s^i$ for $i = 1, \dots, k$.
- If $S \subseteq W$ is an embedded codimension- k submanifold and p is a point of S such that $T_p S$ is complementary to the span of $(V_1|_p, \dots, V_k|_p)$, then the coordinates can also be chosen such that $S \cap U$ is the slice defined by $s^1 = \dots = s^k = 0$.